



Kinematics: Constant Acceleration

• Mark Scheme

Mechanics Chapter 2: Kinematics: Constant Acceleration. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Displacement, velocity, speed and acceleration

- 1 $a = \frac{25-5}{8} \text{ M1}; = 2.5 \text{ ms}^{-2} \text{ A1.}$
- 2 $a = \frac{8-20}{4} = -3 \text{ ms}^{-2} \text{ B1; deceleration } 3 \text{ ms}^{-2} \text{ B1.}$
- 3 (a) $\frac{600}{80} = 7.5 \text{ ms}^{-1} \text{ B1; (b) } 200 \text{ m east B1; (c) } \frac{200}{80} = 2.5 \text{ ms}^{-1} \text{ east B1.}$
- 4 (a) $0.5 \text{ ms}^{-2} \text{ B1; (b) } 36 \text{ s B1; (c) velocity has a direction, speed does not B1.}$
- 5(a) $v = 4 + 0.5 \times 10 \text{ M1}; = 9 \text{ ms}^{-1} \text{ A1.}$
- 5(b) $9 \div 6 = 1.5 \text{ ms}^{-2} \text{ B1.}$
- 5(c) Stage 1: $\frac{1}{2}(4+9)(10) = 65 \text{ m; stage 2: } \frac{1}{2}(9)(6) = 27 \text{ m M1 A1; average speed } \frac{92}{16} = 5.75 \text{ ms}^{-1} \text{ A1.}$

Section B – The suvat equations

- 6 $v = 3 + 2 \times 5 \text{ M1}; = 13 \text{ ms}^{-1} \text{ A1.}$
- 7 $v^2 = 0 + 2(1.5)(48) = 144, v = 12 \text{ ms}^{-1} \text{ M1 A1; } t = 12 \div 1.5 = 8 \text{ s A1.}$
- 8 $0 = 400 + 2a(50) \text{ M1; } a = -4, \text{ deceleration } 4 \text{ ms}^{-2} \text{ A1; } t = 5 \text{ s A1.}$
- 9 (a) $s = 6(8) + \frac{1}{2}(2)(64) = 112 \text{ m M1 A1; } v = 6 + 16 = 22 \text{ ms}^{-1} \text{ A1. (b) } v^2 = 36 + 2(2)(56) = 260, v = 16.1 \text{ ms}^{-1} \text{ B1.}$
- 10(a) $s_P = 15t, s_Q = 1.25t^2 \text{ M1; } 1.25t^2 = 15t \text{ M1; } t = 12 \text{ s A1.}$
- 10(b) 180 m B1.
- 10(c) $2.5 \times 12 = 30 \text{ ms}^{-1} \text{ B1.}$
- 10(d) $\text{Gap} = 15t - 1.25t^2, \text{ greatest when the speeds are equal M1; } 2.5t = 15 \Rightarrow t = 6 \text{ A1; gap} = 90 - 45 = 45 \text{ m A1.}$

Section C – Graphs

- 11 (a) Gradient $\frac{20}{10} = 2 \text{ ms}^{-2} \text{ B1; (b) area } \frac{1}{2}(4+24)(10) = 140 \text{ m B1.}$
- 12 (a) Trapezium shape with correct values marked B1; (b) $\frac{1}{2}(20+10)(12) = 180 \text{ m B1; (c) } 2 \text{ ms}^{-2} \text{ B1.}$
- 13 Top speed $15 \text{ ms}^{-1} \text{ B1; area} = 37.5 + 300 + 75 \text{ M1; } = 412.5 \text{ m A1.}$
- 14 Area: $10T + 1200 + 200 = 1900 \text{ M1 A1; } T = 50 \text{ A1; acceleration } \frac{20}{50} = 0.4 \text{ ms}^{-2} \text{ B1.}$
- 15(a) Trapezium with times 10, 40, 55 B1; V marked B1.
- 15(b) $\frac{1}{2}(10)V + 30V + \frac{1}{2}(15)V = 1275 \text{ M1 A1; } 42.5V = 1275 \Rightarrow V = 30 \text{ A1.}$
- 15(c) Above 20 ms^{-1} for the last $\frac{10}{3}$ s of stage 1 and the first 5 s of stage 3 M1 A1; total $\frac{10}{3} + 30 + 5 = 38.3 \text{ s A1.}$

Section D – Vertical motion under gravity

- 16 $20 = 4.9t^2 \text{ M1; } t = 2.02 \text{ s A1.}$
- 17 $0 = 14.7^2 - 2(9.8)s \text{ M1; } s = 11.0 \text{ m A1; } t = 1.5 \text{ s B1.}$
- 18 $v^2 = 25 + 2(9.8)(30) = 613 \text{ M1; } v = 24.8 \text{ ms}^{-1} \text{ A1; } t = \frac{24.76-5}{9.8} = 2.02 \text{ s A1.}$



19 $15 = 20t - 4.9t^2$ *M1*; $4.9t^2 - 20t + 15 = 0$ *A1*; $t = 0.990$ or 3.091 *M1*; time = 2.10 s *A1*.

20(a) $0 = 21^2 - 2(9.8)s \Rightarrow s = 22.5$ m *M1*; height = 50.5 m *A1*.

20(b) Taking up as positive: $-28 = 21t - 4.9t^2$ *M1 A1*; $t = 5.35$ s *A1*.

20(c) $v^2 = 21^2 + 2(9.8)(28) = 989.8$ *M1*; $v = 31.5$ ms⁻¹ *A1*.

20(d) E.g. air resistance is ignored / the stone is a particle / g is constant *B1*.

Roar-level challenge

21 $s_A = 45 - 4.9t^2$, $s_B = 21t - 4.9t^2$ *M1 A1*; meet when $21t = 45$, $t = 2.14$ s *A1*; height $45 - 4.9(2.143)^2 = 22.5$ m *A1*;
 $v_B = 21 - 9.8(2.143) = 0$, so B is at rest (at its highest point) *B1*.